Kevin Tuttle

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Department of Mathematics
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Education

University of Arkansas
Ph.D., Mathematics, 2022–Current
Fields: Low Dimensional Topology, Contact and Symplectic Topology
M.S., Mathematics, 2022

Brigham Young University
B.A., Mathematics, 2020

Dissertation

"Algorithmic Legendrian Realization of Open Book Decompositions of Contact 3-manifolds"

My upcoming dissertation delineates the relationship between planar open book decompositions of 3-manifolds and the corresponding surgery descriptions of the manifolds. An algorithmic process is achievable to convert from monodromy descriptions of 3-manifolds, and Gompf normal form of the corresponding surgery diagrams. This allows diagrammatic computation of invariants such as Legendrian contact homology. Work completed under Dr. Jeremy Van Horn-Morris, and defense expected May 2027.

Teaching

Mathematics, University of Arkansas
Instructor, Finite Mathematics, Fall 2026
Instructor, Calculus 2, Summer 2026
Instructor, Survey of Calculus, Spring 2026
Qualifying Exam Tutor, Algebraic Topology, Summer 2025
Instructor, Finite Mathematics, Spring 2023–Fall 2025
Instructor, College Algebra, Fall 2022
Recitation Section Instructor, Calculus 1, Fall 2020–Spring 2022

Service

Arkansas Math Discovery Day Booth, "Platonic Solids and Gaussian Curvature," March 2025 and 2026
Mathematics Department Graduate Student Advisory Board, Jan. 2024–May 2025
Mathematics Department Inclusive Excellence Committee, Jan. 2023–Current
Arkansas Prospective Graduate Student Peer Mentor, April 2025
Boy Scouts of America Eagle Scout, May 2017

Research Talks

"Legendrian Realization of Open Book Decomposition Monodromy in Contact 3-manifolds"

Redbud Topology Conference, University of Arkansas, November 7, 2026

AMS Sectional Meeting, Special Session on Contact and Symplectic Topology in Low Dimensions, October 3, 2026

University of Arkansas Topology Seminar, October 1, 2026

"From Integral Curves to Contact Structures"

University of Arkansas Graduate Student Colloquium, January 28 and October 28, 2025

"Symplectic Manifolds and Lefschetz Fibrations"

University of Arkansas Topology Seminar, April 3, 2025

"From Corks to Contact Structures"

South Central Topology Conference IV, March 9, 2025

"Picturing the 4th Dimension: A Topological Journey through Kirby Calculus"

University of Arkansas Graduate Student Colloquium, October 3 and 10, 2024 (two-part talk)

Teaching and Outreach Talks

"Knots and Links: An Exploratory Introduction"

Northeastern State University Undergraduate Math Seminar, November 4, 2026

"Teaching from a Graduate Perspective"

University of Arkansas, Math 5100V (Graduate Teaching Training), September 22, 2026

Active Learning in the Classroom (session co-leader)

University of Arkansas Teaching and Learning Workshop, October 21, 2025

Computing Skills

Python, C++, Mathematica, SnapPy, LaTeX

References

Dr. Jeremy Van Horn-Morris
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Dr. Brian Rickard
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Dr. Matthew Clay
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